

A convergence of TAME (Levin 2022) and Active Inference FEP perspective

57:53

all right welcome

back here with Alexi tolinski a friend
and colleague and collaborator on this
paper temporal depth in a coherent self
and in depersonalization theoretical
model so go for

[Music]

it well uh I'd like to say hello to
everyone and thank you for being with us
after such a long and fruitful day at uh
6: p.m. on on the east coast and uh it
say I appreciate your interest and thank
you Daniel thank you active inference
Institute for this fantastic Symposium
um so uh uh we'll be working talking
today about this paper on dissociative
States and it's a collective effort uh
this paper is in peer review right now
uh early uh preprint was posted uh on
the site and it was uh a work by Michael
Levan Chris Fields Ianot D Costa Rachel
Murphy Daniel who is here uh David
Pinkus and

myself and um this is the link to the
paper if you'd like to take a look at
the earlyer print uh perhaps some of you
would like maybe to get a feel for what
we're talking about and this is not the
patients account this is a a poem uh
from uh Kurt V and his song Pretty
pumpkin I woke up this morning didn't
recognize the man in the mirror then I
laughed and I said oh s me that's just
me then I proceeded to brush some
strangers teeth but they were my teeth

and I was weightless just quivering like
some Leaf come in the window of a
restroom um and this uh poem was cited
by Paul moleno he's a specialist on did
in England on the Psychiatry and
Psychotherapy podcast you can also find
patients account on YouTube I couldn't
post the video here for copyright
reasons um so dissociations are a
spectrum and they range from mild to
severe transient uh to Chronic they can
be benign and pathological and certainly
they're not a bright line category uh
the causes vary for them as well we call
it iology in the clinical World they can
be caused by psychological trauma
neurological conditions substances
seizures anesthesia and uh different
sleep phases there's a subset of
conditions here we call dissociative
disorders typically in the history of
these patients we see prolonged
elevation of stress accompanied by
repeated traumatic experiences in
circumstances where they feel There Is
No Escape typically childhood abuse and
neglect and we call that complex trauma
which has a different presentation than
acute trauma and we'll touch upon why
this specific pathway of iology uh is is
important for dissociative
disorders um
well I'd like to say why uh this matters
what's at stake what's the motivation
for this paper and we have many authors
and they have variable motivations but
I'm a clinician I see patients all day
and um uh for me the main reason for

that is these conditions are hard to
diagnose many patients go undiagnosed
for a long time they're very hard to
treat uh and we're sometimes content
with partial results and the risk of
self harm or suicide is higher for these
patients and there are many many reasons
for that uh we think that perhaps one of
the reasons is that there is no unified
theory of dissociations we have kind of
um neurobiological ideas the corticol
limic inhibition hypothesis we have uh
psychodynamic ideas we have uh attempts
to model dissociations from first
principles by anuna and other colleagues
but uh these different attempts are not
tied together and certainly not tied
together with the patients uh subjective
experiences so we are trying to make
steps in that direction and we don't
claim that we've built a unified theory
of dissociations but what we're doing
here is we're focusing on one aspect of
dissociations one focal point that is a
collapse of the temporal depth in any
kind of dissociation regardless of the
cause by temporal depth we mean how far
into the future the person can plan
what we're hoping for is that this uh uh
focused approach has the potential to uh
allow us to to to um perhaps develop uh
more accurate diagnostic uh uh
procedures and and to possibly um help
increase efficacy in in therapy with
full respect to how difficult complex
and messy this work is um uh additional
point of convergence is we explore
multiple theoretical perspectives in the

paper and they all converge on the same
um points and here is the main thesis um
the collapse of the temporal depth is
necessary and sufficient for the onset
of a dissociative episode so we're going
a little bit further than colleagues who
already mentioned a relationship between
temporal depth and
associations Dean and Anna yunik and
others and Carl friston were saying it's
causal that there is no dissociation
without the temporal depth collapse and
the collapse of the temporal depth will
reliably lead to dissociative episode
both we are um uh saying that temporal
depth collapse can be conceptualized as
a common pathway for the onset of
dissociative symptoms regardless of
itology and finally we suggest that the
focus on restoring the temporal depth
can be an effective strategy in
therapeutic
interventions one of the theoretical
perspectives we explored in the paper is
Michael 11's te framework technological
approach to mind everywhere which I
think is very useful uh and
unfortunately I haven't seen many
uh attempts to uh use it in clinical
applications this is the title and the
link to the paper uh there's so much in
Michael Len's work that is useful uh his
uh paper on cancer which is based on his
experiments is also one of The
Inspirations for our paper here but
here's another inspiration if you dive
into Mike's work you'll probably see uh
what he claims based on experiments that

all intelligences are collective
intelligences however ever at our
subjective level at the level of the
conscious mind we don't feel that way uh
we perceive unity all our percepts
Sensations thoughts feelings come
together in a cohesive and coherent
whole a single conscious experience in
the here and now and uh we will talk
about this seeming uh Paradox uh if you
wish in in our paper um some of the core
tenants of the team framework not all of
them but some key points uh commitment
to graduality the there is no bright
lines uh Michael suggests that
intelligence is built gradually and uh
by accumulation of complexity material
Independence uh our model is not based
on neurons or central nervous system we
will talk certainly about a subset you
know I'm a clinician I work with human
beings so we'll talk about brain-based
models but generally the model applies
to synthetic agents and cells that are
not neurons commitment to empirical uh
research as opposed to just exchange of
opinions on the subject um Dynamics the
C are not fixed they're malleable and
plastic and cell can be nested and
overlapping cooperating and competing
both laterally and across
levels uh this is an illustration of one
of the key Concepts in Len's framework
um uh called a light cone and what you
see here depicted is various organisms
in the middle of the slide you see a
tick and a dog a human being possible AI
by compound intelligence such as an end

colony and you see this sort of nested and embedded structure on the picture on the right and L cone has two uh Dimensions there's a spatial Dimension and a on the horizontal plane and the vertical Dimension is temporal and TI for example is not particularly concerned uh with things happening a mile away uh it cannot recall far from the past and plan far into the future well human beings are capable of uh planning things until uh uh that happen after we die a living will and uh with the use of language and culture and books we can uh use things that happened before we were born so we're capable of huge light cones and I'm highlighting this word capable of it doesn't mean that in our wakeful hours alert hours we are walking around with the significant light cones they contract and expand depending on various circumstances but we're certainly capable of them while ticks or not and the vertical dimension of a lyone is exactly what we mentioned before that is the temporal depth there is of course a perspective on the same thing in fap and I like this very Su phras by Carl friston who said because consequences follow causes the generative model acquires a temporal depth and more formally uh some agents are only capable of modeling their immediate environment and in the Here and Now While others are capable of planning actions into the future and the generative models capable of planning

are referred to as temporally deep generative models so temporal depth under fep is the length of the temporal Horizon that is considered during planning uh we're moving on to describing the if you wish healthy self um and we do mention that it is a materially independent um structure where you know in principle we can talk about of the selves of synthetic agents and you know cell collectives uh uh selves depend on an organism's ability to infer and here we will uh provide uh a viewpoint on the two clinical terms familiar to clinicians such as derealization and depersonalization so the organism's beliefs about the world are the inferences about the external environment and the Continuum of issues in this process we will call derealization while the organism's beliefs about its own internal milieu its body and mind we will call the self and the Continuum of issues of this process we will call depersonalization the S are hierarchical composite nested and with boundaries uh so we consider self to be a system consisting of hierarchically structured beliefs which effectively makes it a dynamic hierarchical generative model and taken in its entirety the self is informationally separated from the external environment by a mark of blanket or near Mark of blanket by boundary would like to emphasize I apologize if this is Trivial to the

audience of the active inference
Institute that this boundary is not
Material like skin it is
informational uh through which the self
and not self interact so to emphasize
the self as we have defined it is a
model it is not an organism the organism
constructs and implements it so the self
is quite literally the construction of
the mind this is an illustration of that
from early work by Michael Lan I think
this was his doctoral dissertation
materials and what you see here is um
Duc embos that started from oite and
then you have blastoderm you after cell
division you have a collection of cells
and at the top left if nothing is done
then this blastoderm will develop into a
single ducky you see this green shrimpy
looking thing that's a single
duck however if you introduce a scratch
in this blastoderm and the wound heals
the same exact cell Collective May
develop
into twins or triplets you see the red
yellow and green duckies now think about
that for a moment what happened
here um this Collective of cells
computed what is me and what is not me
and the hardware did not change but in
one case it made a decision that this is
all me and the rest of it is not me and
another one it says there's three mes so
you can see that this is not
Hardware um
s are also as I mentioned hierarchical
composite nested and they have
boundaries so now we're moving on to

components of the self uh that are each component is separated by their own boundary and Mark of blanket and because we're using the word hierarchical uh we will uh Define that there is a core self a core component if you'd like a visual metaphor that Mark SS uses we can think about an onion and there's different levels of the layers of the on and there's a core uh in the middle we will talk about that as applied to humans and mammals and rodents so now we are in organic uh in organisms uh this term is from effective Neuroscience from Yak pip and Lucid Bin's book um and also you know I believe SS and teral used it uh so core self is non-reflexive non-reflexive and dominated by raw feelings uh and part of purely effective core form of Consciousness so the effective homeostatic mechanisms allow an animal to problem solve in novel environments and I'll give you an illustration from Mark SS um to maybe just um bring it home uh imagine that I'm in the room full of carbon monoxide and I've never been in that before uh so I don't really know what to do but I feel badly I feel badly in a very specific way there's air hunger Suffocation distress uh and let's say I'm moving around the room but if I move to the left and I feel a little bit better and that's information and if I move to the right and I feel worse that's also

information perhaps there's a door on the left and there's a draft underneath the door and I feel better so this Compass this internal mechanism the homeostatic mechanism allowed me to find out a possible solution and um we have a whole collection of these and to mention you know the actual hunger uh for food and thirst and social homeostats such as rage and fear and panic separation distress Etc so the collection of that is what punp and SS conceptualized as the core self uh uh therefore the set of predictions in the core self that all the life sustaining effects will be at or near their settling points which means in emotional terms contentment all my needs are met I am a okay thank you very much right and that is the minimum of the variational free energy if there are deviations from it such as hunger or thirst or uh air hunger then that would be a prediction error accompanied by a negative feeling like in the example that I used above uh in mammals we're not putting the equal sign but we're describing the anatomical structures that are uh participating in in uh serving this function we're talking about the upper brain stem regions including the reticular activating system and the periductal gray they mediate the function of the core self and you can see a lot more details in Mark's book Hidden Spring a journey to the source of Consciousness so um or self is thought

to be present in animals without the cortex such as decorticated cats that are actually more effective and not less and the cases of human children born without a cortex hydrolic children they have emotions when you know uh you put a brother on on the girl's you know uh chest she smile smes and she cries so the core self does not require a functional

neocortex the higher levels of the brain structures in humans including the vortex mediate the higher levels of both Consciousness and the self and sometimes it's referred to as extended

Consciousness which allows us to reflect on what's going on such as to say I feel anxious right

now uh there is an a symmetry in this uh um model where the higher levels the peripheral levels of of the self and and Consciousness cannot exist without the core um while their verse is not true the core can exist without the higher levels one of the other illustrations uh of that which PM cites this is based on Fisher's work in 2016 is that a tiny lesion a 2 cubic millimeter lesion in a parabal nucleus will reliably and IR irreversibly reduce a coma the patient will never become conscious ever again while you will never have anything like that happen in the cortex you know this small lesion in the cortex will not result in catastrophic outcomes like that and this is one of the reasons why neurosurgeons don't usually like to operate around the brain stem this is

too consequential too
risky so everything was said about
Consciousness the core and extended is
applied to self as well that peripheral
self components cannot function without
the operational core self and we will
illustrate that with specific examples
in humans later on and addition to this
onoff where the thing is completely
operational or not we can talk about the
influence and the core self has the
ability to change the regime of
functioning in other more peripheral
self components by inducing phase
transitions in them and the Brain
mechanism for that is generalized
arousal um when the higher levels of
consciousness are present and functional
then we have representations such as
fruit and vegetables and you know in the
core self we have Raw effect such as I
don't know separation distress or hunger
but with the higher levels of
consciousness presence we are capable of
uh uh thinking things like I want an
apple or the affect the drive is in
Freudian language and in SS language is
attached to the
object and similarly we can have these
constructions about the meta object such
as I uh such as I look pale or I am a
pessimist and I here is a mental object
and therefore I is a metacognitive
construct an abstraction and it only
exists at the higher levels of the self
such as autobiographical self it is not
used or even needed in the core self to
reiterate core self as a system has the

capacity to

detect uh the effective prediction

errors and uh attempt to minimize the

variational free energy through action

without any

eye we're moving on to peripheral self

components and we'll describe only a few

uh and the terminology here is different

from what you may read in wonderful book

by Neil Seth being you so the first one

is autobiographical self and that's a

system dynamically representing our

life's history if you for example

remember your 10th birthday and the cake

is on the table and Mom is there and

your friends are there and the sun is

shining and everyone's smiling and it's

Tuesday that's an episode that's an

episodic memory and a collection of them

organized through time is our sort of

life

narrative um there is a bodily self

which is a system dynamically building

inferences about our body and various

representations and interceptive

processing there's a social self which

is a system representing inferences

about how we are seen by others and how

we present ourselves and act in the

social environment each peripheral self

component such as bodily self for

example has its own Mark of blanket and

each component is embedded into the

whole self and also contains

subcomponents creating a nested

architecture that you saw in the T light

cone picture

before about the unity that in humans at

the higher levels of the self such as at the autobiographical self level uh uh uh these these levels create an impression of a unified experience and this is just the experience of the autobiographical self not the entire system the autobiographical self claims to itself and others to be the entire self while it is not so Anil Seth refers to these things as delusion we have a delusion of being a unified I don't know alexity right but you know what we're saying is that it is that system that is deluded um uh the autobiographical self subsystem I'm ambivalent about using the term diluted I think it it's not the optimal one because we use it in the clinical world and I prefer it uh called inference or belief that we're unified State differently we can say that a stable belief that I'm whole in the um I'm complete in the autobiographical self um its generative model contributes to us feeling as a unitary self therefore the subjectively perceived Unity the coherence of the self is an inference related to coherence another thing we can talk about is continuity and nearly is a theology stemming from the definition of Mark of blanket our self remain the same persistent time while all the processes or Communications across its Mark of blanket remain functional and we can um offer you another viewpoint on the same exact phenomenon from if you wish more

neurobiological perspective or
psychological
perspective if you recall maybe you're
at the train station or you know
anywhere in the city and you see someone
and you think you recognize them but
you're not sure you know this sensation
of recognition of familiar uh so the
familiarity novelty calculations is what
we're talking about here and
depersonalization is a breakdown in that
something that is familiar myself
suddenly abruptly becomes very novel and
not
recognized but actually these
calculations neurobiologically are not
uh bright line dichotomous categories
they are probabilistic and they are
Spectrum but we course grain them in the
mind often into yes or no familiar or
novel so that subjective impression of
yes I know this person or not is just
the course grading of the continuous
variable representational capacity is
important uh the Deep generative model
implies that the agent is capable of
planning into the future which in turn
requires an ability to generate store
and retrieve counterfactual data
generate store and retrieve as
memory so therefore when the
representational capacity is completely
or partially impaired for any reason the
agent completely or partially loses an
ability to infer leading to
depersonalization and der
realization if we talk about health and
pathology briefly the two key terms here

are voluntary and involuntary you know
again all of us during our day we can
either sit down to plan things you know
for you know 10 year temporal Horizon
which expands our temporal uh depth or
when we uh meditate or uh rock climb you
know then we can be more in the moment
in the here and
now but uh trauma doesn't ask per I mean
perpetrators are you know there's
nothing voluntary about it and so that's
an involuntary collapse of the temporal
depth uh which usually is leading to
pathological
conditions this is a perspective from
the nonlinear dynamical systems you know
if again I am in uh you know adult
healthy alert functioning and I may have
small deviations from I IM me say I'm
daydreaming or something like that and
the collection of these points from
which I rely ably return to ime can be
described as the attractor Basin and the
whole thing in terms of attractor
landscape is the set point attractor
regime ime and ime would be the
variational free energy minimum and that
is a fairly stable system perhaps in the
autobiographical sum right now in order
to for that system to change some energy
uh needs to be applied and with external
interference of sufficient power this
attractor landscape can change when that
power is applied temporarily there's a
phase transition the attractor landscape
evolves into something else perhaps
during that transition you have a chotic
regime and if we are in the case of

trauma then the functioning can be
chotic for a while but then it may
settle down so to speak into one of the
possible post-traumatic presentations
there may be multiple postraumatic
presentations and one of the possible
options is depicted here this is not
again Universal but you know it's one of
the possible options what you see here
is a multiple coexisting uh Point
attractors represented by local Minima
of the variation free
energy and they can correspond to Alters
in D

ID uh and in this example you also see
that they're surrounded by a global
minimum variation of free energy now
also if you look at the uh area the
region between the local Minima then in
dynamic IAL systems that's called a
repeller and uh repellers are uh
notoriously chaotic so uh that suggest
that you may not have very good
predictability of how you will
transition from alra X to alra y or alra
z uh to say a few more words about
this when we switch between the altars
then we are in regime of itinerancy this
is the term Carl Frist likes to use a
lot and you know justifiably so also
when we have transitions from X Ultra X
to Y that the time is disrupted you know
in the idea you don't have this flowing
continuity of time there's a series of
disruptions effectively then your uh
coherent large light cone is broken into
uh pieces
here now imagine that this patient with

with this condition reached out for help and they want to get better but this uh attractor landscape already acquired some stability and so again we need some interference to destabilize the regime the difference here would be that in therapy in the process of healing the this uh destabilization of the pathological attractor landscape is voluntary controlled and guided ideally by a licensed and competent mental health professional and uh that's that's the uh the difference from the onset of uh acute trauma so Psychotherapy can be once such intervention and it's the treatment of choice we have very limited efficacy with medications and now it's very uh uh uh kind of um fashionable to talk about augmentation with psychedelic treatment or perhaps neuros stimulation it's important to say that just destabilizing that landscape doesn't guarantee anything at all so when people present this idea that you just go to like I don't know and have some iasa or anything ketamine infusion and you're better is a myth because you know the important part is that the patient settles back into health and just the destabilization of this landscape doesn't lead to that you still need uh therapy you still need you know to process the trauma not just to destabilize the pathological regime um and that therapeutic process when you destabilize the the pathological attractive landscape is another phase transition temporarily

temporarily chaotic with the idea that eventually it leads you to a healthier functioning either a attractive landscape corresponding to cohesive uh system or at least a little more coherent system with less Alters partial Improvement let's talk a little bit more about the phase transition so before things happen in in in health we have a nested and embedded uh self and that can be described in in FB is a hierarchy of coupled interconnected attractors car friston you know uh wrote that slower attractor can possibly control the phase transition of the faster attractor and in our system the core self is is feeling based it's effective and it's one of the components uh of uh effects of feelings is generalized arousal and we know that the CH changes in generalized arousal level can lead to the phase transitions of the entire neurocortex from periodic to chaotic State and back so this is one of the possible uh kind of you know uh explanation in our model how this works the phase transitions um you know a few more uh um thoughts about it trauma is not a um guided transition it's an effective storm it's a flood of cortisol that is abrupt and severe and that uh abrupt increase in generalized arousal leading to the destabilization of a healthy attractor landscape in the autobiographical self but then when there's an onset of a dissociative

syndrome after trauma then uh the regime settles down to a lower energy uh perhaps effective numbness that patients talk about and in a a Psychotherapy not immediately but when the patient is ready which can take a year or more you know when there's active trauma work then there is again an increase in in energy level with with the idea eventually to settle down to a healthier regime of

functioning a few clinical observations starting from dissociative disorders which are conceptualized traditionally as uh defensive response to prolonged and inescapable

trauma clinicians talk about uh and patients report this this quality of emotional flatness or deadness or numbing sometimes they call it freeze so in our model it corresponds to issues with Communications between the core self and the peripheral self components for example that autobiographical self in this regime operates as if it is uninformed by the vital emotional informational flows that reaching in the core self one of the possible ways it can be achieved not the only way but one of them is the frontal prefrontal cortex inhibiting the lyic structures that's the cortical lyic inhibition hypothesis and in fep terms that means lowering of precision associated with effective prediction error

messages um we're moving on to also severe dissociations but temporary such

as general anesthesia propofol and uh uh
uh it's a very profound change in the body
that influences the Gap Junctions
between the
cells and recovery from anesthesia is a
very interesting process Michael Lan is
fascinated with it it's actually a
little bit of a miracle that we come
back to be the same person we have been
before and some people don't uh you may
wake up a pirate uh and uh these
conditions are called post-operative
delirium dissociative Amnesia
post-operative cognitive dysfunction
this is also an illustration itself is
computed and plastic and malleable it's
not hard um you know Hardware um and
perhaps we get oriented with the use of
our memory systems including the
episodic memories another severe case is
a seizure where not only
autobiographical self is impaired but
also the social and the bodily self and
one of the reasons for that is the
necessary resources for the functioning
of the self components are not
operational in
seizure uh moderate dissociations uh in
neurological cases but permanent ones
are for example the perhaps one of the
most famous neurological patients hm who
had a bilateral hippocampus damage uh
and uh in Britain there's Clyde wearing
and Anil Seth describes Clyde wearing in
his book being you this patients do not
have episodic memory encoding and
effectively they live in about 10
minutes of the here and now uh there is

no past and no future beyond that and if you read Clyde wearing's diaries that he talks about the ser of Awakenings he said oh I just woke up and 10 minutes later oh I woke up and then I woke up and if you show these Diaries to a clinician who is an expert in did they will which I've done you know I talked to Roxy wolf one of the fellows of the international Society for the study of Trum associations and she said that you know this patient looks perpetually depersonalized there is no continuity itself Corso patients they don't recall the past they can fabulate the past and in Alzheimer's in early phases there's issue with recent autobiographical memories but some access to the remote autobiographical memories all of these cases are discontinuity in autobiographical self um uh more frequent perhaps cases or substances ketamin I doses of THC or iasa lead to depersonalization and dualization and importantly if you read the studies then these patients actually report they tell you that their sense of time is impaired not only that the planning is impaired as well um if you have a alcohol blackout and the person tells you on Wednesday that they have no idea whatsoever what happened on Tuesday after 6 then uh this is a discontinuity this is a gap in episodic memory uh there's also something called traumatic Amnesia there very said cases of high school shootings where you know the person literally doesn't remember

you know 15 minutes of what was happening in the cafeteria what happens in acute PTSD is that you know uh hippocampus is reaching cortisol receptors and there's a flood of cortisol and the effectively the episodic memory of trauma the cohesive context contextual time stamp memory is not laid down in a stable fashion you have shards of glass you have phobic memories semantic memories procedural memories but not the episodic one which is also discontinuity autobiographical self and that is involuntary uh and we talked about mild benign everyday uh dissociations uh next time you wake up you know maybe tomorrow the day after you know try to monitor how what's going on uh and you may catch that pH of what met singer calls you know a non-ic self where things are just flowing through your mind but your eye is not fully out borous yet uh to summarize the level of consciousness regime of Consciousness pathological States and all the underlying resources necessary for the successful operation of each self component collectively influence the complex dynamics of the system um a neurobiological perspective briefly the historically most famous one is uh cortical limic inhibition hypothesis by Sierra and barus who postulated that uh prefrontal cortex is uh inhibiting the lyic structures including the amydala and the interior singular cortex

ACC the subsequent studies don't fully support that hypothesis but partially support it I like the work by Ruth Lenius and her colleagues and guess what you know her uh revision her refinement of the cortical liic inhibition adds Dynamics transience context and itinerancy all the things we like uh uh uh they talk about two regimes even in patients with dissociative syndromes over modulation and under modulation ation again itinerancy in not one regime but multiple regimes over modulation is indeed hypoactive amigdala and hyperactive prefrontal cortex which is described above but also these patients less frequently but but you can see it in a reverse regime where amygdala is active and and PFC is less active uh uh over modulation dominates in dissociative uh conditions they refined the anatomical Network and talked about different sub regions of amygdala the central medial amygdala basolateral amygdala dorsolateral periductal gray ventral lateral periductal gray Thalamus and other regions and this network of regions collectively Works to create specific post traumatic regimes they also described uh the regime where patient has access to painful memories usually uh is accompanied by under modulation very active state with amigdala active and uh when they don't have access to painful memories they're more in a dissociative State we can also briefly

talk about what are the anatomical
substrates and the neuro psychological
substrates of the temporal depth
collapse and again acute trauma hpc
compal damage and dementia are affect
the episodic memory
system uh if there's an insulin damage
then you can expect the disruption in
the bodily memories and perhaps in
camine use you may have a dysfunction of
prefrontal Cortex and frontoparietal
Network
which renders the work in memory uh of
the patient dysfunctional you can
imagine other circumstances when you
have PFC in front of paral network
impaired that usually leads to a
psychotic state where you know it's very
hard to think you know we need work in
memory to be able to
think um about dissociative disorders
again the point made at the very
beginning that prolonged inescapable
exposure to highly stressful
environments can be seen as lasting
stress beyond the agent's ability to
manage which results in the breakdown of
um tame light into the uh components a
fragmentation decoherence and
discontinuity when the landscape uh
attractor and repell Landscape
corresponding to a pathological
condition takes root and in contrast to
that if you look at the single uh
episode of camine intoxication then you
have transient working memory disruption
leading to transient
dissociation again to summarize lasting

or temporary severe or mild dissociative experiences are accompanied by the collapse of the self temporal depth and we are ready for discussion which I very much look forward

to thank you Lexi great

presentation thank

you

okay where should we begin what would be

interesting or or given where the work

has gotten over the last few

weeks where what do you feel like is

most

relevant well perhaps any colleagues

have comments I don't know if Jus is

with us and he's a esteemed neuros

psychiatrist and he sees a lot of

patients and perhaps neurological

patients and he can comment on some of

the things we've mentioned I I had a uh

clinical meeting with with uh colleagues

who do a lot of dissociative works and I

mean I'm seeking feedback and critique

and so whatever ever people have

whatever thoughts people have yeah if

there's anyone in live chat they can

write a question a few pieces that stuck

out differently in how you presented it

uh in comparison from working on the

paper was the narrative self's role in

the

maintenance of

The Narrative of

unity and about what is the experiencer

of that

or is that all it can do why does it do

that why does the autobiographical self

have that kind of a voice or a cognitive

function is audio working okay can yeah
can we hear each other I I think why is
a tricky question you know it might also
be tell logical question I would say
that it's probably functional you know
uh for us to have coherence elves uh and
um I don't know why uh but
um we are observing it so we have
descriptive level we have observations
that I feel like the same Elixir and
it's a little bit of a miracle that as
we know that the cells in the organisms
they are they die and they're born but I
still feel like the the same person
even so uh we change developmentally I'm
not the same person as I have been at
age 5 or age 15 but I still maintain
this story if you wish this believe that
I am the same and um I also protect the
internal milieu from external threats you
know of the same organism and if we look
at other species say an oak tree that
you know uh grows from an acorn it is
still the same Oak it's the same organ
uh so um I I perhaps in biology as well
there's a need uh for the system inside
the mark of blanket to maintain that
saying whatever is inside the mark of
blanket is the same
me but it is a belief you know and uh
um I don't know if I'm answering your
question maybe I'm just free associating
to it so yeah is that a clinically valid
method um well it it reminds me of
mortal computing and there's a lot of
focus on how behavioral activities in a
niche relate to the material stability
and I think you really drew carefully on

Mike Levan at all's work to talk about
the analgous

Dynamics on a landscape of a
narrative self and that's just kind of
that was like a trampoline but then you
also complicated that with multiple
elves like the social and the narrative
and so then there are these
multiple

disassociative or like it's kind of they
each have their
own

unity and yet they can be discordant and
unified and so then is there any general
framework for different possibilities
open-endedly of different regimes
these are great questions or even maybe
challenges to develop it further but I
agree with you if we look in the bodily
self right I don't perceive my hand to
be a separate entity I believe I I feel
for it to be part of the whole organism
myself my body right and so in the
bodily self there's also a level of
unity I'm just saying that this is again
a computation within that specific sub
system that my body is one whole self
and we can see how that gets uh perhaps
distorted in somatoparaphrenia or even
rubber hand illusion where uh this hand
is perceived as not you know so uh and
so we see that the system is plastic and
malleable and and prone to change but
and when when no uh interference is
happening it continues to be cohesive is
one body and one mind you know this
autobiographical I I I've noticed you Ed
the term narrative uh self I think that

Anil Seth's term and I prefer autobiographical for the reason that uh rats they have episodic memories and they don't have our verbal abilities um uh and so they they also have episodic memories which are what we when the contextual memories right and I think that uh there's the same eye in all autobiographical memories and that's unity uh yeah yeah thanks you a few it's it's it it probably is more accurate to say it's a narrative cognitive capacity that encompasses autobiographical or Alo biographical type narratives um and I'm reminded of uh NPR radio discussion from a few days ago that I listened about Memoir and about what Memoir was and could it only be written when the person did or didn't change a lot but what or who doesn't change in every single one of these ways so then what would that mean like what so that that that was kind of a funny connection and then and then um do you have any thoughts on that or have another active comment no I'm just enjoying your comment I don't know if I have a a I thought on that yeah you you've see many clinical Memoirs and presentations of self- understandings and all of this so there's just so many and it gets brought out so sparsely like and indirectly and it makes a lot of sense to the person who might be saying it but

that that could you could be too or not
confident enough about
that interesting you brought it up thank
you for that Daniel I think that you
know when we theorize say in Dynamic
theories then we uh usually think about
different sub agencies of the mind like
Freud for example in his know structural
model he had the Eid ego and super ego
where I want pistachio Asam is the Eid
and pistachio es is fat and calories
that's the super ego response and stash
esum is \$4 and you have three and a half
is the ego uh but there's a whole bunch
of them we can name a sort of a sub
agency of an internal judge you know
which is sort of super English or a
neutral Observer or uh an accountant
who's running Itali or
and all of this so we all have sort of a
jury there of different kind of regimes
of operation if you wish but uh the
patient is often surprised because when
they for example self- criticize and you
talk about that in therapy and say do
you notice the voice of your judge it's
kind of you know uh not very fair I mean
there's no misdemeanor everything's a
felony and judge is present all the time
like no it's not it's me I'm doing it
you know I screwed up you know so again
you know they default regime is is me
myself and I that I am you know I am bad
and when we we discuss these things and
we open it up and we show the different
kind of uh we show a theater with
different voices or a jury with
different representations it's usually

kind of an Insight it's a surprise but that allows a bit more control um for the patient in clinical work soon okay I'll read a question in the live chat Pablo FM wrote thank you very much I find it super interesting I wonder why humans seek for drug experiences having been so decisive in human history coffee could be a great example tea tobacco alcohol have also been very popular among civilizations it's a great question I don't know if I have a good answer it's a huge question and there's a good paper you can find it on Google's scholar a paper by me zelner Mark SS and other colleagues I think Yak pip was alive it's called why depression feels bad and what do addicts really want and it's based on effective neuroscience and what it postulates is that uh of course there are individuals who have multiple drugs that they are you know using all the time so there's an upper and the downer and other things but there are usually some innate kind of predispositions where people who are wound up in tense they sometimes crave alcohol because it's uh binding to the Gaba receptors and it's sort of you know sedating us a little bit it makes you a little more less kind of wired up people who are very painfully lonely and uh uh detached and they crave human uh comfort and you know to feel together and loved and accepted they may if they discover opioids that would be a you know lock and key match that would be a dangerous

combination um and uh perhaps when somebody's oriented to Performance and they're very bored all the time and they want to something you know to do better and more they may gravitate toward uh stimulants and so there's these things uh and there's some common factors among different agents uh such as you know I believe that multiple substances they have the dopaminergic effect and you know a lot of motivations are subjective in individuals I've had patients say that I'm smoking pot to slow down time I've had patients tell me that you know p is my oldest friend I've been doing it long longer than any person that I've been with uh some people seek this state of numbness and they want to kind of not feel as acutely they want to sort of feel desensitized so these states of dissociation can be desired um and that's briefly I guess what I what I wanted to say about it so well thank you I'm reminded of a comment from ma in the live chat earlier this morning and she was kind of pointing to how one mode of how we've seen active inference applied in these presentations is looking like athology at the sequences of behavior but not really looking at the self-maintenance as a behavior like especially talking about digital agents or or that it's just about what do they perceive what do they do it's taken as a granted or a separate question how they exist yet also there's this

self-maintenance

self-reconstruction

character that doesn't even need to have
anything to do with the athology its

Dynamics can just be like

morphologically playing

out not necessarily doing some cognitive
task out in the world in any

particularly effective

way yeah it's a very interesting thought

and for some reason I'll allow myself

the Liberty to free associate again it

reminded me of yal Harari who went

interviewed he said you guys go to the

gym You observe your diet and you take

care of your body and he said I take

care of my mind I meditate an hour a day

and uh it goes to silent Retreat a month

and a year and the interviewer

challenged him I think he was from The

Daily Show he said how do you manage

that and her said I got no kids you know

so but in addition to that he's also in

therapy so there's a uh that idea of

really maintaining the mind and not just

the body with behaviors we're all

oriented to what do I do do do what do I

think think think which are sort of

verbal virtual behaviors but how do you

feel as my great colleague Raji uh

Warrior says we're human beings not

human doings we have a very tough time

being we're all about doing more that's

true and so uh yeah and also Harari

talks about informational hygiene where

a lot of stuff in the social media is

noise and trash and a lot of it is

consumed and we have pollution but you

know so he doesn't do you know he reads books and he thinks in Millennia but I don't know I mean I went sideways from your comment but it's just that's what where my thought went so yeah it it seemed like a very relevance when people are discussing the the the unity or the articulation of Mind and Body which is kind of what this unified promise on the cover of the textbook suggests is what what would it be to have a unified model of brain mind behavior and so on then the the analogy or the connection with the physical and the cognitive is just it has to be taken as a starting point not like a hot take yeah yeah but what you just said reminded me when I read Michael Len's work and got more familiar with it I actually had regret I'm like why were we not taught that that was a bit of a revolution in my mind it was a big pivot because if you go through graduate program in psychology it's all about the unity you know I am the same you know cohesive and coherent and the fact that we have intelligence at the level of individual cells and then the cell Collective and then tissue and organ and all of this thing that I think is one montic Alexi is a pyramid with Communications bottom up and top down and laterally is is a revolution I think that you know I I wonder what models of psychology can be built with that and they don't exist yet

it's all about one big block and what happens to it I am depressed you know there's no collectives this is one big block

right yeah it's a really important insight I think that's the kind of frictions and and experiences will start to learn more about and see produce in different ways like just seeing how you connected the topics to specific quotes translated the ontology re worded

descriptions using kind of like a translation sometimes really just as simple as a lookup of words just authors using one word and being able to see an ontology term equivalent or roughly related it's uh and

and what what does that look like or I guess as your last comments what do you think after to the paper after this now like what are the directions towards which people would say okay that's applying active inference in a clinical setting I my goal eventually I think is to get to Therapeutics and for example you know um actually I probably will not use that example it's it's a good material for the second paper that we'll follow up with uh but I I do think that you know the question is what do we do in therapy differently how do we understand how do we formulate cases differently with this information I I haven't heard it clearly said out loud in the did literature there's a lot of

perspectives there's a lot of endless
complexity and there's a lot of nuance
but if we can highlight a central point
and say there is a temporal depth
collapse I think it allows us to do
something with it in clinical work and
in diagnostic work and that's my hope it
may not materialize but this is what I'm
hoping for so
well even from a
researcher's
ethological perspective it's kind of
like that's the transferable and the
interesting research question how could
you be
observing not on the inside and make
some kind of inference or assessment
that would matter clinically with no map
territory pretense not that it's 100%
reliability just taking a clear look at
Diagnostics available rethinking it from
dynamical systems
perspective maybe different uh
Biometrics maybe pupil it's like it's
not it's just kind of that's the open
question and then where it provides
clinical utility
that's where it
matters yeah thank you perfectly said
yeah I agree okay thank you
Alexi see you thank you Daniel see you
byebye e